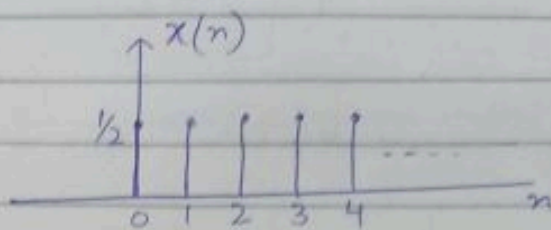


PROBLEMS :- DTFT :-

①

(i) $x(n) = d^n u(n)$; $d < 1$

Suppose $d = 1/2$



$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \\ &= \sum_{n=0}^{\infty} d^n u(n) e^{-j\omega n} \end{aligned}$$

We know ; $\sum_{n=0}^{\infty} d^n = \frac{1}{1-d}$

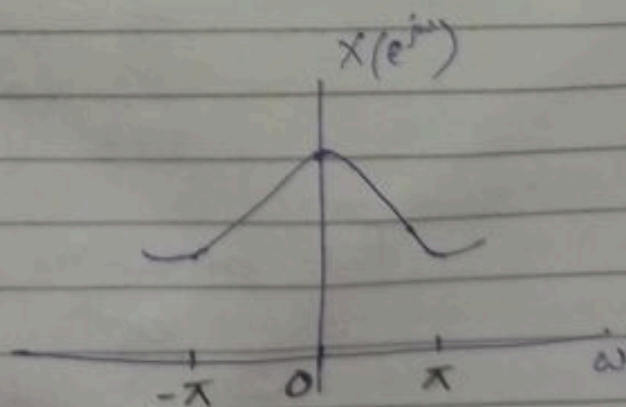
$$= \sum_{n=0}^{\infty} d^n e^{-j\omega n}$$

$$= \sum_{n=0}^{\infty} (d e^{-j\omega})^n$$

$$X(e^{j\omega}) = \frac{1}{1 - d e^{-j\omega}}$$

when $\underline{\omega = 0}$

$$\begin{aligned} X(e^{j\omega}) &= \frac{1}{1 - d e^0} = \frac{1}{1-d} \\ &= \frac{1}{1 - 1/2} = 2 \end{aligned}$$

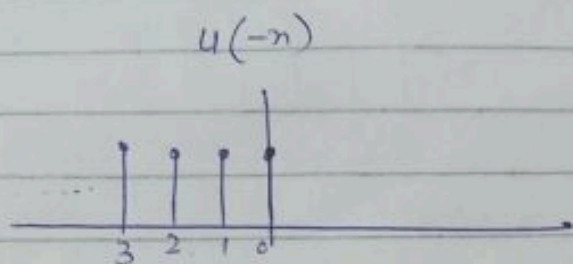
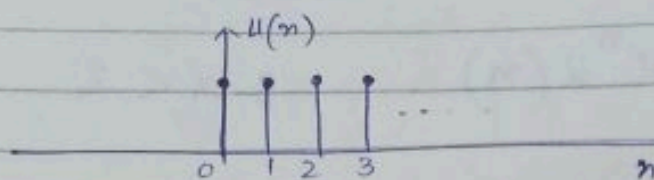


$$\omega = \pi \quad \frac{1}{1 - d e^{-j\pi}} = \frac{1}{1 + d} = \frac{2}{3}$$

(2)

$$x(n) = 2^n u(-n)$$

(2)



$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=-\infty}^{\infty} 2^n u(-n) e^{-j\omega n}$$

$$= \sum_{n=-\infty}^0 2^n e^{-j\omega n}$$

$$= \sum_{n=0}^{-\infty} 2^n e^{-j\omega n}$$

$$= \sum_{n=0}^{-\infty} (2e^{-j\omega})^n$$

Put $n = -n$

$$-n = 0 \Rightarrow n = 0$$

$$-n = -\infty \Rightarrow n = \infty$$

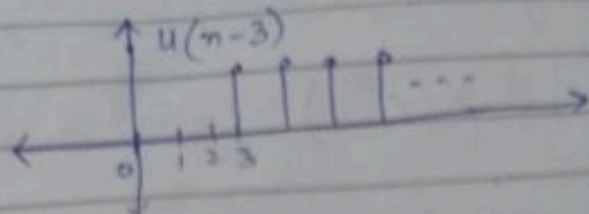
$$= \sum_{n=0}^{\infty} (2e^{-j\omega})^{-n}$$

$$= \sum_{n=0}^{\infty} (2^{-1} e^{j\omega})^n = \frac{1}{1 - \frac{1}{2} e^{j\omega}}$$

$$X(e^{j\omega}) = \frac{1}{1 - \frac{1}{2} e^{j\omega}}$$

$$(3) \quad x(n) = \left(\frac{1}{3}\right)^n u(n-3)$$

(3)



Method 1

$$\begin{aligned} X(e^{j\omega}) &= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \\ &= \sum_{n=3}^{\infty} \left(\frac{1}{3}\right)^n u(n-3) e^{-j\omega n} \\ &= \sum_{n=3}^{\infty} \left(\frac{1}{3} e^{j\omega}\right)^n \end{aligned}$$

$$n=3 \Rightarrow n-3=m$$

$$\text{If } n=3 \text{ then } m=0$$

$$n=\infty \text{ then } \infty-3=m$$

$$\infty = m$$

$$= \sum_{m=0}^{\infty} \left(\frac{1}{3} e^{-j\omega}\right)^{m+3}$$

$$n-3=m$$

$$n=m+3$$

$$= \sum_{m=0}^{\infty} \left(\frac{1}{3} e^{-j\omega}\right)^m \left(\frac{1}{3} e^{-j\omega}\right)^3$$

$$= \left(\frac{1}{3} e^{-j\omega}\right)^3 \sum_{m=0}^{\infty} \left(\frac{1}{3} e^{-j\omega}\right)^m$$

$$= \frac{1 \cdot e^{-j3\omega}}{27} \cdot \frac{1}{1 - \frac{1}{3} e^{-j\omega}}$$

$$X(e^{j\omega}) = \frac{e^{-j3\omega}}{27 \left(1 - \frac{1}{3} e^{-j\omega}\right)}$$

(4)

Method 2: Using Properties :-

$$x(n) \xleftrightarrow{\text{DTFT}} X(e^{j\omega})$$

$$x(n-n_0) \xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega})$$

$$\Rightarrow x(n) = \left(\frac{1}{3}\right)^n u(n-3)$$

$$\left(\frac{1}{3}\right)^3 \left(\frac{1}{3}\right)^{n-3} u(n-3) \leftrightarrow e^{-j\omega 3} \left(\frac{1}{1 - \frac{1}{3}e^{-j\omega}}\right) \left(\frac{1}{27}\right)$$

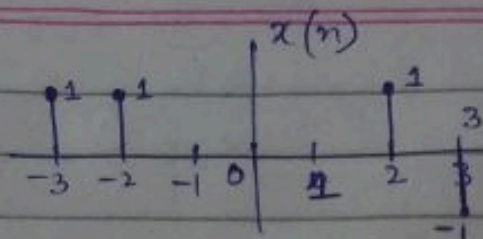
$$\text{FT of } \left(\frac{1}{3}\right)^n u(n) \leftrightarrow \frac{1}{1 - \frac{1}{3}e^{-j\omega}}$$

Shift by 3 steps

$$\left(\frac{1}{3}\right)^{n-3} u(n-3) \leftrightarrow \frac{e^{-j\omega 3}}{1 - \frac{1}{3}e^{-j\omega}}$$

$$\left(\frac{1}{3}\right)^3 \left(\frac{1}{3}\right)^{n-3} u(n-3) \leftrightarrow \frac{1}{27} \frac{e^{-j\omega 3}}{1 - \frac{1}{3}e^{-j\omega}}$$

(4) :-



(5)

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= x(-3) e^{-j\omega(-3)} + x(-2) e^{-j\omega(-2)} + x(1) e^{-j\omega(1)} + x(2) e^{-j\omega(2)} + x(3) e^{-j\omega(3)}$$

$$= (1) e^{3j\omega} + (1) e^{2j\omega} + (1) e^{-j\omega} + (-1) e^{-2j\omega} + (3) e^{-3j\omega}$$

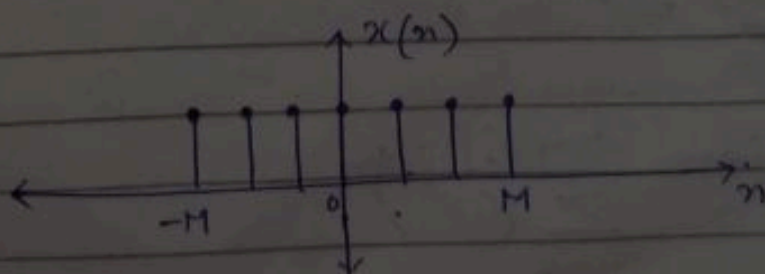
$$= e^{3j\omega} - e^{-3j\omega} + e^{2j\omega} - e^{-2j\omega}$$

$$= 2j \left(\frac{e^{3j\omega} - e^{-3j\omega}}{2j} \right) + 2 \left(\frac{e^{2j\omega} - e^{-2j\omega}}{2} \right)$$

$$X(e^{j\omega}) = 2j \sin 3\omega + 2 \cos 2\omega$$

(5) :-

$$x(n) = \begin{cases} 1 & ; |n| \leq M \\ 0 & ; |n| > M \end{cases}$$



$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$= \sum_{n=-M}^M (1) e^{-j\omega n}$$

$$= \sum_{n=-M}^M (e^{-j\omega})^n \quad \text{--- ①} \quad (6)$$

We know;

$$\sum_{n=0}^{\infty} d^n = \frac{1}{1-d}$$

$$\sum_{n=0}^N d^n = \frac{1-d^{N+1}}{1-d}$$

Put $n+M=m \longrightarrow n=m-M$

when $n=-M \quad m=0$
 $n=M \quad m=2M$

Eq ① will become

$$= \sum_{m=0}^{2M} (e^{-j\omega})^{m-M}$$

$$= \sum_{m=0}^{2M} e^{-j\omega m} \cdot e^{+j\omega M}$$

$$= e^{+j\omega M} \sum_{m=0}^{2M} (e^{-j\omega})^m$$

$$= e^{j\omega M} \left(\frac{1 - (e^{-j\omega})^{2M+1}}{1 - e^{-j\omega}} \right)$$

$$= e^{j\omega M} \left(\frac{1 - e^{-j\omega(2M+1)}}{1 - e^{-j\omega}} \right) \quad (7)$$

If I take half of exponential out.

$$= e^{j\omega M} \frac{e^{-j\frac{\omega(2M+1)}{2}}}{e^{-j\frac{\omega}{2}}} \left(\frac{e^{j\frac{\omega(2M+1)}{2}} - e^{-j\frac{\omega(2M+1)}{2}}}{e^{j\frac{\omega}{2}} - e^{-j\frac{\omega}{2}}} \right)$$

$$\underbrace{\qquad\qquad\qquad}_{\Downarrow}$$

$$\frac{e^{j\omega M} \cdot e^{-j\omega M} \cdot e^{-j\frac{\omega}{2}}}{e^{-j\frac{\omega}{2}}}$$

Divide Numerator & Denominator by $2j$

$$\frac{\sin \frac{\omega}{2} (2M+1)}{\sin \left(\frac{\omega}{2} \right)}$$

$$\therefore X(e^{j\omega}) = \frac{\sin \left(\frac{\omega}{2} (2M+1) \right)}{\sin \left(\frac{\omega}{2} \right)}$$

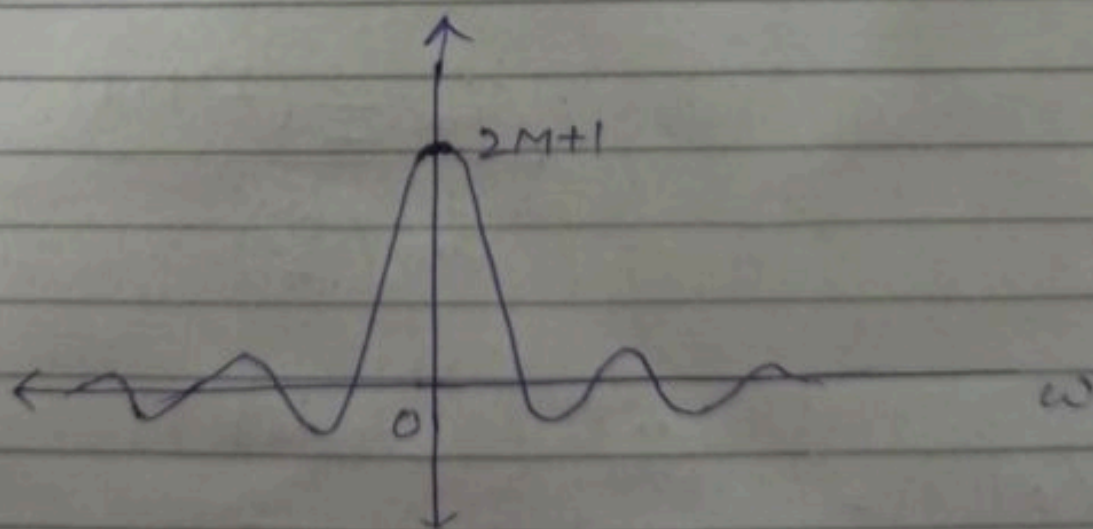
$$X(e^{j\omega}) = \frac{\sin\left(\frac{\omega}{2}(2M+1)\right)}{\sin\left(\frac{\omega}{2}\right)}$$

L'Hospital's Rule

$$\frac{\cos\left(\frac{\omega}{2}(2M+1)\right) \left(\frac{2M+1}{2}\right)}{\cos\left(\frac{\omega}{2}\right) \left(\frac{1}{2}\right)}$$

$$\omega = 0$$

$$X(e^{j0}) = 2M+1$$



IDFT

9

① :-
$$X(e^{j\omega}) = \frac{6 - \frac{2}{3}e^{-j\omega} - \frac{1}{6}e^{-2j\omega}}{-\frac{1}{6}e^{-j2\omega} + \frac{1}{6}e^{-j\omega} + 1}$$

find $x(n)$.

Sol :-

Put $e^{-j\omega} = t$

$$= \frac{6 - \frac{2}{3}t - \frac{1}{6}t^2}{-\frac{1}{6}t^2 + \frac{1}{6}t + 1}$$

$$= \frac{-\frac{1}{6}t^2 - \frac{2}{3}t + 6}{-\frac{1}{6}t^2 + \frac{t}{6} + 1}$$

$$= \frac{\frac{1}{6}t^2 + \frac{2}{3}t - 6}{\frac{1}{6}t^2 + \frac{t}{6} + 1}$$

$$= \frac{6\left(\frac{t^2}{6} + \frac{2}{3}t - 6\right)}{6\left(\frac{t^2}{6} + \frac{t}{6} + 1\right)}$$

$$= \frac{t^2 + 4t - 36}{t^2 + t - 6}$$

$$\frac{1}{t^2 - t - 6} = \frac{1}{t^2 + 4t - 36}$$

$$\frac{1}{t^2 - t - 6} = \frac{1}{(t-3)(t+2)}$$

$$\frac{5t - 30}{t^2 - t - 6}$$

$$= 1 + \frac{5t - 30}{t^2 - t - 6}$$

$$= 1 + \frac{5t - 30}{(t-3)(t+2)}$$

$$X(e^{j\omega}) = 1 - \frac{A}{t-3} + \frac{B}{t+2}$$

$$= 1 - \frac{3}{t-3} + \frac{8}{t+2}$$

$$= 1 - \frac{3}{e^{-j\omega} - 3} + \frac{8}{e^{-j\omega} + 2}$$

$$= 1 + \frac{1}{1 - \frac{e^{-j\omega}}{3}} + \frac{4}{1 - (-\frac{1}{2})e^{-j\omega}}$$

Taking IDTFT

$$= \delta(n) + \left(\frac{1}{3}\right)^n u(n) + 4 \left(-\frac{1}{2}\right)^n u(n)$$